Repeat-purchase prediction: pipeline overview

Homework — Group 34 · what each stage does (left) and why (right)

1. Data

train.csv: 6,534 purchases (2015–2025), 56% repeat
test.csv: 986 later purchases (2025–2026)
clientes.csv: optional customer table

why

Test lies after train in time, and most test customers already appear in train

2. Cleaning (fixed rules)

65 empty training rows dropped (none in test)
negatives, 1e9 sentinels, inflated means and max > total spend → missing
prior mean refilled from total / count

why

The same errors exist in test: row-level rules learn nothing from the data, so they are applied to train and test alike

3. Validation design

random 5×2 folds vs grouped vs time-ordered, each checked on the 2024–25 rows
→ 5×2 folds to compare options
→ final check: fit ≤ 2023, score 2024–25

why

5×2 has the smallest standard error, but 38% of its training labels see validation rows (90-day target): its scores are optimistic

chosen on the 5×2 folds

4. Preprocessing recipe (fitted per fold)

fill: median (numbers) / own level (labels)
log1p on every numeric column
target encoding + min-max scaling
(winsorizing and ordinal encoding rejected)

why

log1p: +0.010 F1 on 9/10 folds; target encoding ties one-hot with 34 columns instead of 119 → the simpler option

5. New features and clientes.csv

history flags, recency_ratio, purchases_per_year, recent_share_90d, margin and discount share
+ country, region, customer_since_year

why

All 11 together: +0.0015 F1 (inside the noise), none harmful → passed to feature selection; customer_id and random_noise never used

6. Feature selection (2-of-3 vote)

filter: point-biserial relevance, redundancy 0.8
wrapper: RFE, k = 40 chosen on validation
embedded: L1 logistic regression (C = 0.1)
→ kept if 2 of the 3 agree, refitted per fold

why

Same F1 as all 45 columns with about 31; alone, L1 and the 3-stage sequence lose
→ 33 columns on all training rows

7. Model

logistic regression, default settings (C = 1)
grid of 20 (L1/L2, C, class weight): no gain
decision threshold 0.425 for F1

why

Threshold: +0.011 F1 on 10/10 folds; trades balanced accuracy for recall, because the competition metric is F1

8. Submission

refit on all training rows, predict test.csv
ID unchanged, sample_submission order
→ submission.csv (70.6% predicted repeat)

why

test.csv only goes through transform: no fit ever sees the test rows

Assessment

Every step is refitted on the training part of each fold.
Paired fold-by-fold comparisons: 'better' beyond 2 SE.
Metrics: F1 on the repeat class, ROC-AUC, bal. accuracy.
Final check: model and threshold refitted on rows up to 2023, then scored on the 2024–25 rows.

Results

2024–25 check: F1 0.782

floor (always 'repeat') 0.754
5×2 folds: F1 0.755, ROC-AUC 0.788 (floor 0.718)